

Contributor Dataset QA Guide

AI-assisted review process for building a practice-ready and model-training-ready VCE Mathematics dataset

Purpose

The contributor role is **Dataset Reviewer + AI-assisted Dataset Improver**. The goal is not to click approve quickly. The goal is to make every question safe for student practice now and useful as high-quality training data later.

Rule: AI can suggest improvements, but the original source PDF and human mathematical verification remain the source of truth.

What Contributors Must Produce

Quality level	Required result
Practice ready	Question text matches the source, answer and accepted forms are correct, marks and topics are correct, and the student marker works or the record is clearly marked manual review.
Training ready	Everything in practice ready, plus a clear worked solution, useful marking rubric, common mistakes, accepted answer variants, reviewer notes, and completed checklist.
Expert review	Use this when AI or the contributor improves the record but there is remaining uncertainty about mathematical correctness, marking rules, or whether the item can be safely auto-marked.

Contributor Workflow

- 1 Open **Dataset QA**. Enter your reviewer name before saving any decision.
- 2 Select the correct dataset source, for example **2025 Mathematical Methods Exam 1**.
- 3 Open the first **Ready for QA** record in the queue.
- 4 Open the original exam PDF or trusted source. Compare the rendered question and all math notation.
- 5 Use AI to challenge or improve the worked solution, accepted answers, marking rubric, topic labels, and common mistakes.
- 6 Verify AI output yourself against the source and the mathematics. Do not approve only because AI says it is correct.
- 7 For auto-check questions, run the marker using the expected answer and confirm the score is correct.
- 8 Complete the training quality checklist and choose the training readiness level.
- 9 Choose Approve, Needs Fix, Manual, or Reject. Add a short note explaining exactly what you checked.
- 10 Move to the next record and repeat until the assigned queue is complete.

Fields To Check In Dataset QA

Field	What to check	Expected contributor action
Dataset source	Correct exam year, paper, or practice-topic dataset.	Select the source that matches the source PDF or assigned batch.
Reviewer name	The system requires a reviewer name before saving.	Enter your name once before making decisions.
Question preview	Rendered math, wording, question number, diagrams if present, and line breaks.	Compare with the original PDF. Fix markdown if the display is wrong.
Question markdown	Raw editable question text.	Correct OCR/Mathpix errors, missing brackets, wrong symbols, or unclear formatting.
Topic and subtopic	Learning classification used for analytics, practice routing, and future model training.	Check against the question skill being tested. Use AI as a second opinion, then verify.
Correct answer	Canonical expected answer.	Must be mathematically correct, exact where required, and in a form accepted by the marker.
Accepted answers	Equivalent forms students might reasonably type.	Use AI to suggest variants, then test or verify equivalence before saving.
Worked solution	Reasoning path, not just final answer.	Improve clarity, check all algebra/calculus/probability steps, and remove unverified AI claims.
Marking rubric	How marks are awarded.	Keep each criterion clear and mark-linked. It should explain what earns each mark.
Common mistakes	Likely student errors useful for feedback and model training.	Use AI to brainstorm, then keep only plausible mistakes tied to this question.
Answer test	Marker behavior for auto-check records.	Paste the expected answer. Approve only if the marker result matches the expected marks.

AI-Assisted Review Rules

AI should help contributors find gaps and improve training value. AI should not replace source checking.

Use AI for	Do not use AI for
Suggesting alternate accepted answers; checking a worked solution; identifying common mistakes; improving rubric wording; checking topic/subtopic fit; explaining why an item may need manual review.	Approving a record without checking the source; inventing official marks; changing the mathematical answer without verification; adding unsupported solution steps; treating AI output as official VCAA guidance.

Suggested AI prompts

- Check this VCE Mathematical Methods answer and identify equivalent accepted answer forms: [paste question and answer].
- Review this worked solution for mathematical correctness and missing steps: [paste solution].
- Suggest common student mistakes for this question, but only if they are specific to the question.
- Classify this question by topic and subtopic. Explain why the classification fits.
- Turn this marking rubric into clear mark-by-mark criteria without changing the total marks.

Reviewer note rule: if AI was used to improve a field, write it in the reviewer note, for example *AI-assisted solution review used; verified against source and derivative rules*.

QA Decision Rules

Decision	Use when
Approve	The record is practice ready. Required checklist items are complete, source matches, answer is correct, and marker/manual-review behavior is appropriate.
Needs Fix	The record is useful but has a fixable issue, such as OCR error, wrong topic, missing accepted answer, weak solution, or unclear rubric.
Manual	The record is valid but should not be auto-marked, usually because it needs explanation, proof, graph interpretation, or human judgement.
Reject	The record is not suitable, duplicated, badly corrupted, not aligned to the source, or unsafe for student use.

Training Quality Gate

The checklist prevents low-quality data from entering the training set. Approve only after the applicable checks are completed.

Checklist item	Meaning
Source matched	Question text and expected answer match the original source.
Topic checked	Topic and subtopic represent the actual skill tested.
Answer checked	Correct answer is mathematically correct and exact where required.
Accepted answers checked	Equivalent answer variants are valid and not over-permissive.
Marker test passed	Auto-check marker gives the correct score for the expected answer.
Rubric checked	Rubric explains mark allocation clearly.
Solution checked	Worked solution is accurate, clear, and useful for future training.

Training readiness

- **Practice only:** safe for students, but not rich enough for model training yet.
- **Training ready:** source, answer, solution, rubric, accepted answers, common mistakes, and notes are complete and verified.
- **Expert review:** potentially valuable but should be checked by a stronger maths reviewer before training use.

Definition Of Done

- Every assigned record has a saved decision and reviewer note.
- Approved records have no unresolved source, answer, topic, marker, rubric, or solution issue.
- Training-ready records include common mistakes and a verified worked solution.
- Needs Fix records explain the exact fix required.
- Manual records explain why auto-marking is unsafe.
- Rejected records explain why the record should not be used.
- Publish is used only when the batch owner has confirmed the approved records are ready to go live.

Reviewer Note Templates

Situation	Example note
Approved auto-check	Checked against 2025 Exam 1 PDF. Question text and answer match. Marker returns 1/1 for expected answer. Product rule topic is correct. AI-assisted solution review verified.
Needs fix	Source question is correct, but accepted answers are too narrow. Add equivalent form $2x\cos(x)-x^2\sin(x)$ and rerun marker.
Manual review	Question requires explanation/working judgement. Answer alone is insufficient for reliable auto-marking.
Expert review	AI suggested an alternate solution, but the mark allocation is uncertain. Needs expert maths review before training-ready approval.
Reject	Record does not match the source question and cannot be safely repaired from the current data.

Common Pitfalls

- Approving because the question displays on screen without checking source accuracy.
- Trusting AI-generated algebra without independently verifying each step.
- Adding answer variants that are not mathematically equivalent.
- Marking a record training ready when the worked solution or common mistakes are empty.
- Using broad topic labels when a precise subtopic is available.
- Forgetting to run the marker for auto-check records.
- Publishing approved records before the batch owner asks for release.